

A Reinforcement Learning Model of Dopamine-Mediated Appetite Regulation in Health, Parkinson's Disease, and Dopaminergic Treatment

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Introduction

- Appetite is regulated by homeostatic pathways signalling nutritional needs and hedonic pathways evaluating the reward value of food.
- The hedonic pathway, mediated by dopamine, implements reward-based learning to decide whether to eat or not, thereby ensuring daily nutritional needs are met.
- In Parkinson's disease (PD), dopamine loss disrupts this pathway, leading to appetite loss and fat loss-driven weight reduction, while dopaminergic treatment can induce compulsive overeating and weight gain, underscoring dopamine's central role in appetite control.
- This study investigates dopamine's role in appetite regulation under normal conditions and its disruption in PD and dopaminergic treatment states.

Methods

- A cortico-basal ganglia reinforcement-learning model is employed to simulate daily dietary decisions in three conditions: healthy control, PD, and PD under dopaminergic medication.
- Dopamine is modeled as a temporal-difference error signal. Nutritional features (calories, carbohydrates, proteins, fats) and food cues define the state, and actions correspond to eating or not eating.
- The striatum acts as the critic estimating food value, the globus pallidus interna (GPi) mediates action selection, and the subthalamic nucleus (STN) with the globus pallidus externa (GPe) supports exploration.

Model details

- The reinforcement learning model consists of a single agent that learns to regulate its energy by choosing Eat or Not Eat in response to food cues and its current physiological state. $Actions \in \{Eat, Do not Eat\}$
- If the agent decides to *Eat*, it consumes the available food item and assimilates its nutritional energy, updating its physiological state; choosing *Not Eat* leaves the state unchanged.
- The agent is rewarded for maintaining homeostasis, with higher rewards assigned when the physiological state (Φ) approaches a predefined target physiological state (Φ^*)

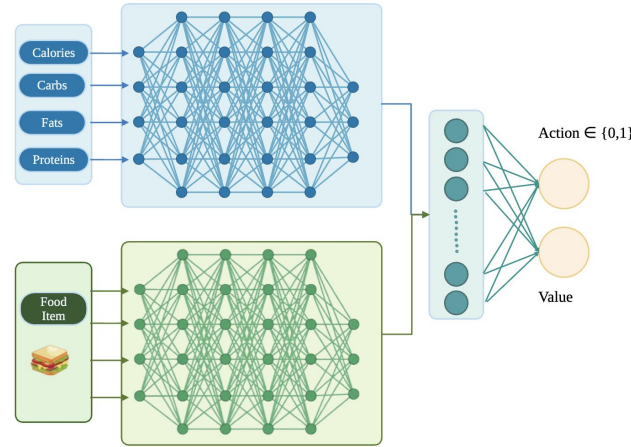


Figure 1: Actor-critic network schematic inspired by cortico-basal ganglia circuitry, illustrating action selection by the actor and value estimation by the critic during dietary decision-making.

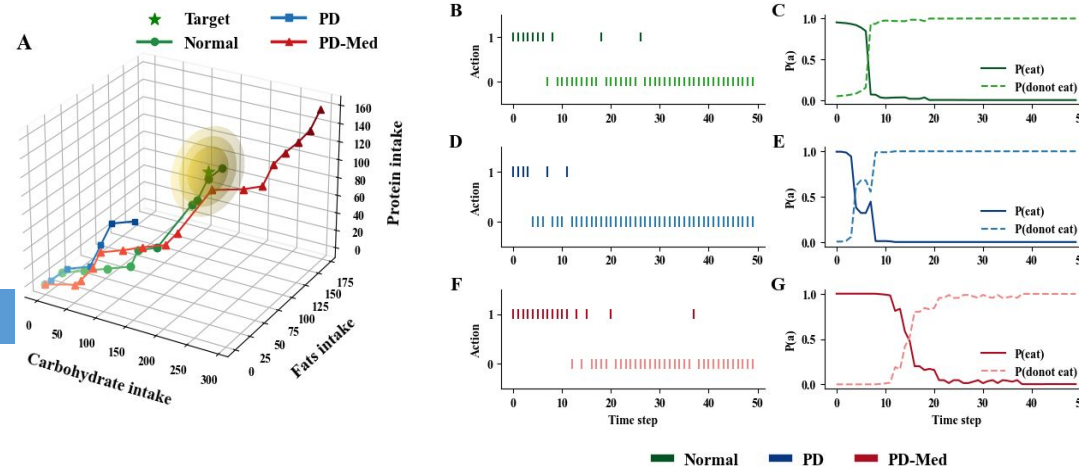


Figure 2: (A) Three-dimensional trajectories of carbohydrate, fat, and protein intake under Normal, PD, and PD with medication (PD-Med) conditions. Each trajectory starts at a light-colored point and gradually darkens over time. The green star indicates the target physiological intake, while graded spheres represent regions of increasing positive reward that guide learning. The physiological state is defined by (calories, carbohydrates, fats, proteins), with target values of (2030, 158, 115, 94.8). Under normal conditions, the model converges to (1930, 159, 115, 87); under PD, to (1115, 113.4, 38, 70); and under PD-Med, to (3045, 293, 171, 159). (B–G) Action selections and corresponding action probabilities over time for each condition, where Action 0 denotes *do not eat* and Action 1 denotes *eat*.

Discounted return:

$$G_t = \sum_{k=t}^T \gamma^{k-t} R_k$$

TD error:

$$\delta_t = R_t + \gamma V(s_{t+1}) - V(s_t)$$

Actor loss:

$$\mathcal{L}_{\text{actor}} = -\sum_t (G_t - V(s_t)) \log \pi(a_t | s_t)$$

Critic loss:

$$\mathcal{L}_{\text{critic}} = \sum_t \delta_t^2$$

PD:

$$\delta_t^{\text{PD}} = \min(\delta_t, \delta_{\text{lim}})$$

PD medication:

$$\delta_t^{\text{MED}} = \delta_t^{\text{PD}} + \delta_{\text{med}}$$

$$R_t = f(d_t); \quad d_t = \|\Phi - \Phi^*\|$$

where rewards are scaled nonlinearly, strongly amplified near the target, and penalized if the agent overshoots the optimal physiological range.

- Learning follows an **actor-critic** reinforcement-learning scheme in which temporal-difference (TD) prediction errors guide updates to both the policy and value estimates.
- In the Parkinsonian model, TD prediction errors are attenuated to represent reduced dopaminergic signaling, limiting learning efficacy.
- Under dopaminergic medication, the magnitude of TD signals is partially restored, improving learning while remaining constrained relative to healthy control.

Results

The simulations match experimental observation:

- Healthy model: Achieves the nutritional balance and reaches closest to the target physiological state
- PD model: Trends toward caloric deficit and under-selection of food.
- PD-medication promotes compulsive, reward-driven overeating and ends up overshooting the target

Conclusions

In conclusion, under intact dopaminergic conditions, the model accurately estimates value and meets daily nutritional needs. Under reduced dopamine, impaired value estimation accelerates the decline in eating probability, leading to underconsumption, whereas dopaminergic medication attenuates this decline, predisposing to overeating.

References

- Keramati, M., & Gutkin, B. (2014). Homeostatic reinforcement learning for integrating reward collection and physiological stability. *Elife*, 3, e04811.
- Chakravarthy, S., Balasubramani, P. P., Mandali, A., Jahanshahi, M., & Moustafa, A. A. (2018). The many facets of dopamine: Toward an integrative theory of the role of dopamine in managing the body's energy resources. *Physiology & behavior*, 195, 128-141.